One State to Draw Them All

A Unified Latent State for Expression, Space, Morphology, and Time in Single-Cell Biology

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Built with Claude — Life Sciences Hackathon (Researcher Track)

Abstract

Single-cell measurements are fragmented across modalities: transcriptomics, morphology, spatial context, and temporal dynamics are acquired on different platforms with no shared coordinate system. We treat a cell as a dynamical system with one hidden state and ask whether a *single* 128-dimensional latent state S—readable, generative, and intervenable—can serve as that coordinate system. To our knowledge this is the first cell model to hold all four axes in one such state. We train one shared encoder $E : \mathbb{R}^{422} \to \mathbb{R}^{128}$ on 200,000 cells from a 10x Xenium human colon-cancer sample, with three heads that decode S back to (i) expression, (ii) a self-supervised morphology embedding, and (iii) spatial neighborhood structure. A classifier-free-guidance (CFG) diffusion decoder then *generates* the cell's microscopy image directly from S, and the *frozen* encoder transfers an independent EMT time-course into the same space. Every axis is validated against a shuffle/identity control: expression R^2=0.574, spatial R^2=0.331, morphology gain +0.30, time Spearman ρ=0.50, and a generation cell-specificity gap of +0.17 at w=3. We report the full method, the equations, and an honest account of the failure (blurry hallucination) that motivated the CFG fix. Code, weights, and a public interactive demo are released.

1 Introduction & Motivation

Biology has always measured a cell one window at a time: a transcriptome in one experiment, a microscopy image in another, a spatial slide in a third, a time-lapse in a fourth—each on a separate platform, each living in a separate distribution, none asked to become the same object. Our thesis is that they already *are* the same object. A cell is a dynamical system with a single hidden state—its regulatory program—and expression, morphology, spatial position, and time are merely four windows onto that one state. If that is true, then the right representation is not four models but *one shared state* S, decodable to every window and rollable through time.

We take this literally and build S into a proper world model in the machine-learning sense: a compact latent you can *read* (decode to any axis), *roll forward* (move through time), *intervene on*, and *decode back into an image*. We draw on the world-model lineage—Ha & Schmidhuber's latent-plus-dynamics formulation, DeepMind's Dreamer (planning by imagination in latent space), and Genie 2 (action-controllable worlds from an autoregressive backbone with a diffusion decoder). **To our knowledge, no prior cell model holds all four axes in one intervenable, generative state, and being first to do so is the core of the contribution.**

Crucially, the novelty is *not* any single block: CFG, DINOv2, GNNs, and FiLM are all established, and we adapt rather than invent them. The novelty is (i) being first to fuse all four biological axes into one cell state that is simultaneously readable, generative, and intervenable, and (ii) proving each axis *real* with an adversarial negative control instead of asserting it—shipping a running, servable system rather than a schematic.

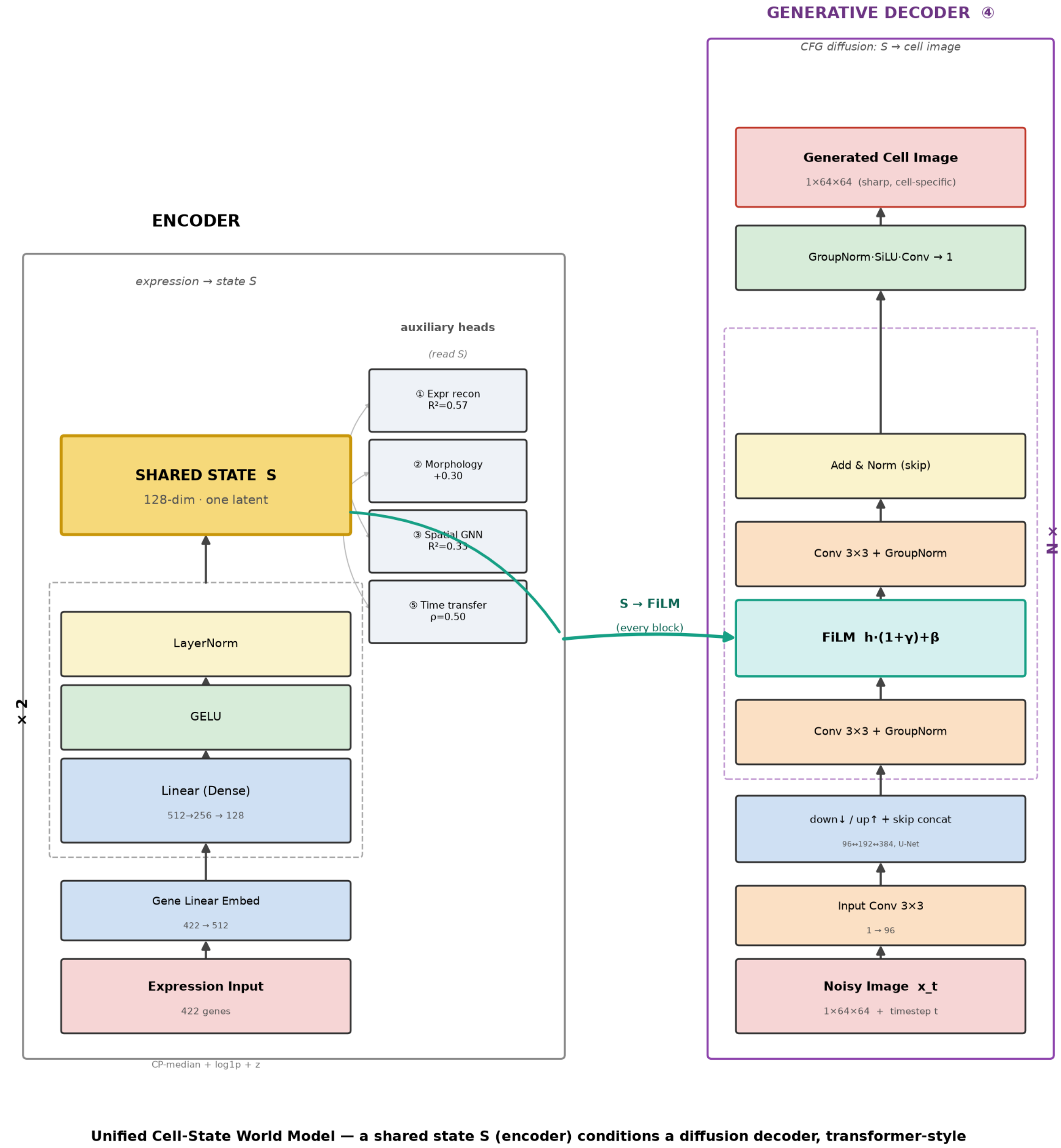


Figure 1: Architecture. A shared encoder maps 422-gene expression to a 128-d state S (left). Three auxiliary heads decode S to expression, morphology (DINOv2-384), and spatial neighborhood; a CFG diffusion U-Net (right) generates the cell image from S via FiLM conditioning in every ResBlock. Time is a *transfer*: the frozen encoder projects an independent EMT time-course into the same S.

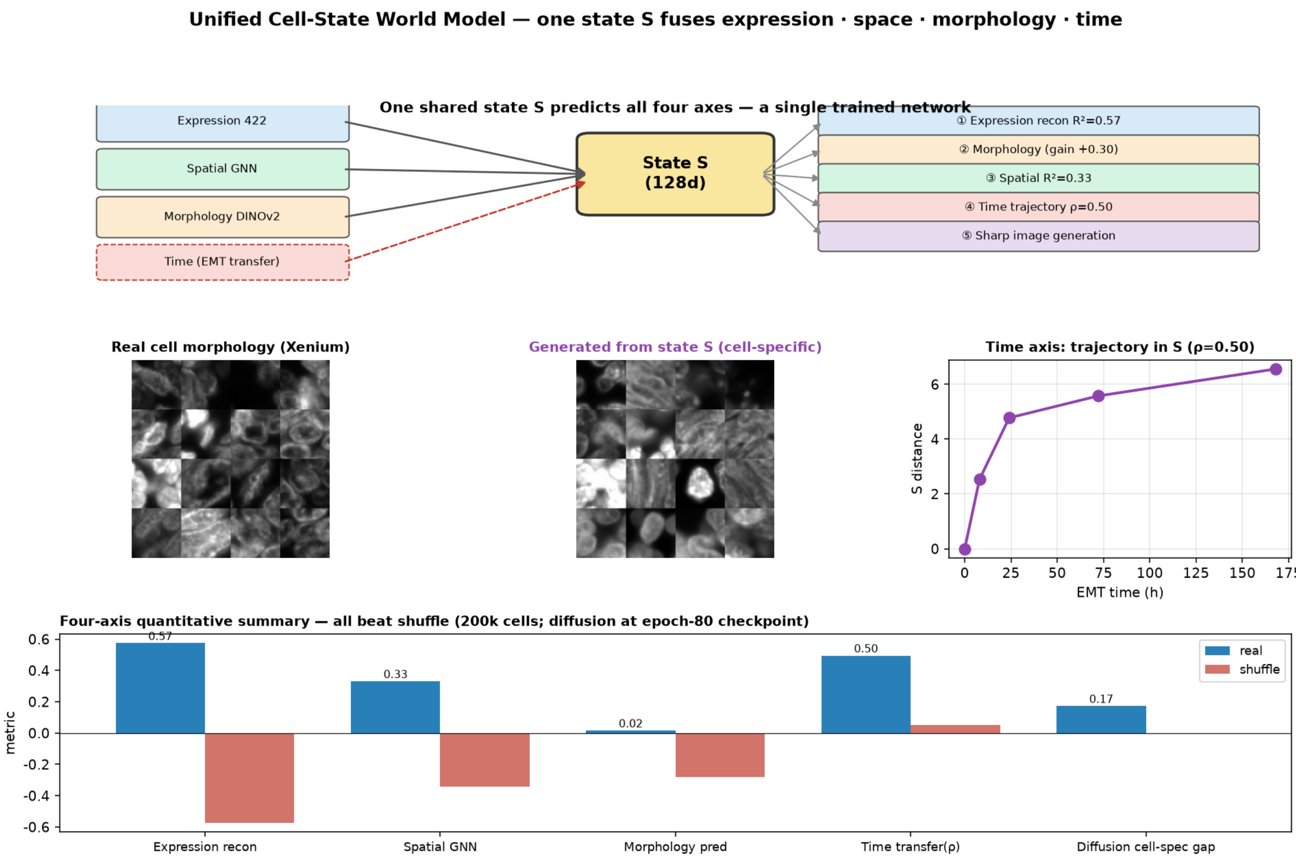


Figure 2: The system at a glance. One shared 128-d state S predicts all four axes (top). Real Xenium cells vs. images generated purely from S (middle-left); the EMT time trajectory in S (middle-right); and the four-axis quantitative summary, every axis beating its shuffle control (bottom). A live public demo lets anyone reproduce the middle-left panel interactively.

2 Related Work

Expression→image generation. GeneFlow (2025) couples an attention-based RNA encoder with a conditional U-Net trained by rectified flow; its authors report FID 20.73 on the same Xenium platform. Spatia (2025) is reported to scale to 49 donors, 17 tissue types and 12 disease states (∼17M cell–gene training pairs), fusing morphology, expression, and space. All external figures in this section and in Table 1 are as reported by the cited works, not re-measured here. On expression→image *alone*, both exceed us in scale and absolute image metrics. **Single-cell latents.** scVI provides a probabilistic negative-binomial latent; our encoder is deterministic and, crucially, shared across modalities. **Our niche.** The four-axis fusion into one state with a control at every step, including a cross-platform *time transfer*, is what we did not find in prior work.

2.1 Where we sit in the Virtual Cell landscape

The "virtual cell" has become one of the most active frontiers in computational biology. The Arc Institute's STATE model is reported to have been trained on observational data from 167M cells and perturbational data from over 100M cells across 70 human cell contexts; its Virtual Cell Challenge (*Cell*, 2025) frames the central task as predicting a cell's transcriptomic response to a genetic perturbation in a held-out cell type. CZI's TranscriptFormer is reported to span 112M cells across 12 species, and its rBio distills such a model into a conversational reasoner. (These scale figures are as reported by the respective groups.) The common thread is decisive: these are, almost without exception, **single-modality (transcriptomic) models whose goal is perturbation→expression prediction**. They embed expression in a latent space and attend over genes; they do not draw the cell, place it in tissue, or move it through time.

Our work is deliberately orthogonal (Table 1). We do not compete on transcriptomic scale—we have 2×10^5 cells against their 10^8. We ask a different question: can one latent state be *simultaneously* an expression code, a morphology code, a spatial code, and a temporal coordinate? Every flagship virtual-cell model answers only the first. Two recent spatial-image works—GeneFlow and Spatia—do generate morphology from expression and exceed us there, but neither unifies all four axes into a single predictable state with a negative control at each one, nor demonstrates a cross-platform temporal transfer. That specific combination is our contribution.

3 Method

3.1 Preprocessing

Raw transcript counts $c \in \mathbb{Z}_{\geq 0}^{422}$ per cell are normalized to a common scale, log-transformed, and standardized per gene:

$$x_g = \frac{\log\left(1 + m\, c_g / \sum_{g'} c_{g'}\right) - \mu_g}{\sigma_g}, \tag{1}$$

where m is the dataset median total count (m=64), and (μ_g, σ_g) are the per-gene mean and standard deviation. The same (μ_g, σ_g, m) are stored and reused verbatim for the time-transfer experiment.

3.2 Shared encoder and auxiliary heads

The encoder is a multilayer perceptron with two GELU/LayerNorm blocks,

$$S = E(x) = W_3\, \phi\big(W_2\, \phi(W_1 x)\big) \in \mathbb{R}^{128}, \tag{2}$$

$\phi(\cdot) = \mathrm{LN}(\mathrm{GELU}(\cdot))$, with widths $422 \to 512 \to 256 \to 128$. Three heads $h_k(S) = U_k\, \mathrm{GELU}(V_k S)$ decode S:

(1) Expression reconstruction. $\hat{x} = h_{\mathrm{expr}}(S)$, loss $\mathcal{L}_{\mathrm{recon}} = \|\hat{x} - x\|_2^2$.

(2) Morphology. The target is a self-supervised embedding of the cell's 64×64 DAPI crop I obtained from a frozen DINOv2 ViT-S/14, $z = \mathrm{DINOv2}(I) \in \mathbb{R}^{384}$. The head regresses it: $\mathcal{L}_{\mathrm{morph}} = \|h_{\mathrm{morph}}(S) - z\|_2^2$. Using a self-supervised target (rather than raw pixels) isolates *shape/texture* statistics from background intensity.

(3) Spatial GNN. Build a k-nearest-neighbor graph (k=12) on cell centroids (x, y). For center cell i with neighbors $\mathcal{N}(i)$, a message-passing layer predicts the center state from the mean neighbor state:

$$\hat{S}_i = g\Big(\tfrac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} S_j\Big), \quad \mathcal{L}_{\mathrm{spatial}} = \|\hat{S}_i - S_i\|_2^2. \tag{3}$$

The encoder and all heads are trained jointly:

$$\mathcal{L}_{\mathrm{joint}} = \mathcal{L}_{\mathrm{recon}} + \mathcal{L}_{\mathrm{morph}} + \tfrac{1}{2}\mathcal{L}_{\mathrm{spatial}}. \tag{4}$$

3.3 S-conditioned CFG diffusion decoder

To *generate* the microscopy image from state, we train a denoising diffusion model conditioned on S. With a cosine noise schedule $\{\beta_t\}_{t=1}^{T}$ (T=400) and $\bar{\alpha}_t = \prod_{s \leq t}(1 - \beta_s)$, the forward process is

$$q(x_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\ (1 - \bar{\alpha}_t)\mathbf{I}\big). \tag{5}$$

A U-Net $\epsilon_\theta(x_t, t, S)$ predicts the noise. Conditioning enters through a vector $c = \mathrm{MLP}_t(\mathrm{emb}(t)) + \mathrm{MLP}_S(S)$, injected by **FiLM** in *every* ResBlock at every resolution:

$$\mathrm{FiLM}(h, c) = h \odot (1 + \gamma) + \beta, \qquad [\gamma, \beta] = W_f\, c. \tag{6}$$

Table 1: Positioning against representative virtual-cell and spatial-generative models. "Axes" = which of {expression E, morphology M, space P, time T} the single model represents. Scale figures are as reported by each work.

Model	Primary task	Axes	Gen. image?	Scale	Neg. control?
Arc STATE	perturbation→expression	E	no	167M obs / 100M pert	benchmark
CZI TranscriptFormer	cross-species expression	E	no	112M cells, 12 sp.	benchmark
scGPT	multi-omics / perturbation	E	no	33M cells	task metrics
GeneFlow (2025)	expression→image	E, M	**yes** (FID 20.73)	Xenium (1 platform)	FID
Spatia (2025)	expression+space+morph	E, M, P	**yes**	∼17M cell–gene pairs	scale
Ours	**unified state, all axes**	**E, M, P, T**	**yes** (gap +0.17)	200k cells	**every axis**

For classifier-free guidance we drop S to a learned null token $\varnothing$ with probability p=0.1 during training, minimizing $\mathbb{E}_{t,x_0,\epsilon}\|\epsilon - \epsilon_\theta(x_t, t, S_{\text{drop}})\|_2^2$. At sampling time we extrapolate:

$$\tilde{\epsilon} = \epsilon_\theta(x_t, t, \varnothing) + w\big(\epsilon_\theta(x_t, t, S) - \epsilon_\theta(x_t, t, \varnothing)\big), \qquad (7)$$

and denoise with 50 DDIM steps, $x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\,\tilde{\epsilon}$, $\hat{x}_0 = (x_t - \sqrt{1-\bar{\alpha}_t}\,\tilde{\epsilon})/\sqrt{\bar{\alpha}_t}$. Weights are tracked with an exponential moving average (decay 0.999).

3.4 The cell-specificity gap (why not FID)

A decoder can look sharp yet ignore its condition—generating a plausible but *generic* cell ("hallucination"). FID cannot detect this. We therefore measure whether a generated image matches *its own* target more than a random one. With generations $\hat{I}_i$ and reals I_i, and a permutation π,

$$\text{gap} = \underbrace{\tfrac{1}{n}\textstyle\sum_i \|\hat{I}_i - I_{\pi(i)}\|^2}_{\text{MSE to shuffled}} - \underbrace{\tfrac{1}{n}\textstyle\sum_i \|\hat{I}_i - I_i\|^2}_{\text{MSE to true}}. \qquad (8)$$

gap > 0 means generation is cell-specific; gap ≈ 0 is exactly the hallucination signature we diagnosed and fixed (§4).

3.5 Time as a transfer, not a fourth training axis

Xenium provides expression, morphology, and space at single-cell resolution but is a static snapshot. To test whether S carries temporal structure, we take an *independent* EMT scRNA-seq time-course (A549 under TGF-β1, $t \in \{0, 8\text{h}, 1\text{d}, 3\text{d}, 7\text{d}\}$), keep the 175 genes shared with the Xenium panel, zero-pad to 422, apply the *stored* Xenium normalization, and project with the *frozen* encoder: $S^{\text{EMT}} = E(x^{\text{EMT}})$. No retraining. We then measure the correlation between real time and the projection of S^{EMT} onto the principal temporal axis.

4 Development History: What We Actually Tried

This project was not a straight line. The path itself is evidence of how deeply the design space was probed, and each turn was forced by a concrete negative result rather than a guess.

(0) We rejected our first success. Following the dominant modern world-model recipe—autoregressive temporal structure plus diffusion spatial generation, as in Genie 2, GameNGen, and DIAMOND—we trained an autoregressive next-frame predictor on live-cell time-lapse. It beat a persistence baseline by +41%, and *we rejected it as a success*: it had only learned the optical flow of pixels and understood nothing about the cell. That honest failure reframed the entire project—the goal was a meaningful *state*, not prettier pixels—and pushed the dynamics off the pixel grid and into a latent.

(1) A naïve time-series world model felt biologically thin. Our next system predicted how a perturbed cell-state *distribution* shifts over time with an optimal-transport transition model (Sinkhorn W_2), beating an identity baseline by +21.5%. It worked—but we judged the biological significance to be weak: predicting a distribution drift is not the same as understanding a cell. This pushed us toward a *state* that could be decoded back into a real cell image.

(2) A deterministic decoder produced correct-but-blurry blobs. Regressing the microscopy image from expression with an MSE decoder beat the mean image by +22% and a shuffled control by +41%—so it captured *something* real—but every output was a soft ellipse. MSE on a many-to-one map (one expression profile, many valid shapes) provably collapses to the conditional mean. We measured this directly: reconstructed variance was only 23% of real. The blur was not a bug to tune away; it was the mathematically correct answer to the wrong objective.

(3) A diffusion decoder was sharp but *hallucinated.* Switching to diffusion made images sharp (sharpness 0.05 > real 0.032), but we caught a subtle failure with our own control: MSE-to-true (0.100) *equaled* MSE-to-shuffled (0.099). The decoder was drawing beautiful, generic cells that ignored the condition. This is precisely why we do not trust FID alone, and why we defined the cell-specificity gap.

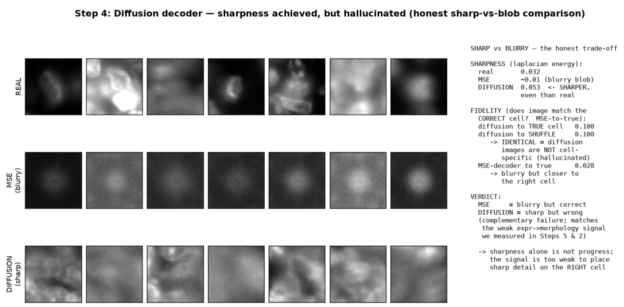


Figure 3: The failure that shaped the design. Real cells (top); the MSE decoder is cell-specific but blurry (middle); plain diffusion is sharp but hallucinated (bottom)—MSE-to-true equals MSE-to-shuffled. This complementary failure motivated CFG.

(4) CFG closed the loop. Classifier-free guidance with FiLM conditioning at every resolution restored cell-specificity: the gap grew monotonically with guidance weight, reaching +0.17 at w=3 while staying sharp. The condition finally mattered.

(5) From five separate models to one shared state. Each axis above had lived in its own network. The final step—and the point of the paper—was to force them through a *single* S: one encoder, jointly decodable to expression, morphology, and space, then generating images from that same S, and finally absorbing time by transfer. Along the way we also solved a mundane but decisive engineering problem: uploading data to the GPU was 100× slower than having the GPU pull the 13 GB source directly from the public CDN (20 MB/s), which is what made 200k-cell training feasible at all.

5 Experimental Setup

5.1 Compute environment

All training was performed on a single **NVIDIA A100-80GB** GPU rented on-demand through Modal, driven programmatically from Claude Science. The container image was CUDA 12.4.1 with Python 3.11, PyTorch 2.x, and mixed-precision (AMP) enabled. Data persisted on a Modal Volume (`xenium-unified`) so that every job—training, checkpointing, sampling—read and wrote the same 1.3 GB preprocessed tensor without re-uploading. Local orchestration ran CPU-only (14 cores, 64 GB RAM); no figure or number in this report required a local GPU.

5.2 Data pipeline

The source is the 10x Genomics Xenium `Human_Colon_Cancer_P1_CRC_Add_on_FFPE` bundle (307,762 cells, 422-gene panel, a 6.5 GB morphology OME-TIFF). Naively uploading the extracted tensors from the local machine to the GPU ran at $\sim$ 0.2 MB/s. We instead had the GPU container *pull the 13 GB source zip directly from the public 10x CDN* at 20.8 MB/s—a 100× speedup that made 200k-cell preprocessing a 4-minute job rather than a multi-hour bottleneck. For each of N=200,000 cells we range-extracted three zip members and produced: (i) a 422-dim expression vector, (ii) a 64×64 DAPI morphology patch centered on the cell, (iii) its (x, y) centroid, and (iv) a DINOv2 ViT-S/14 embedding of the patch. The JPEG2000-compressed OME-TIFF required the `imagecodecs` decoder—a concrete pitfall we hit and record for reproducibility.

5.3 Training configuration

Unified encoder + heads. AdamW (lr=3×10^{-4}, weight decay 10^{-5}), cosine-annealed over 100 epochs, batch size 1024, loss weights $(\lambda_{\text{recon}}, \lambda_{\text{morph}}, \lambda_{\text{spatial}}) = (1, 1, 0.5)$, spatial graph k=12. Total 0.72M parameters; wall-clock $\approx$ 2 minutes.

S-conditioned diffusion. AdamW (3×10^{-4}, weight decay 10^{-4}), cosine schedule, batch size 256, T=400 cosine-β steps, CFG drop p=0.1, EMA decay 0.999, U-Net base width 96, conditioning dimension 256 ($\sim$7M parameters). Sampling uses 50 DDIM steps. To survive pre-emption, the job committed an EMA checkpoint and a progress JSON to the Volume every few epochs; the run converged (loss $\approx$ 0.0233) and was stopped at epoch 80/120 at $\sim$103 s/epoch, with all reported figures drawn from the epoch-80 EMA weights.

Time transfer. No training: the frozen encoder projected 3,133 main-arm EMT cells (GSE147405, A549 TGF-β1) using the 175 shared genes and the stored Xenium normalization.

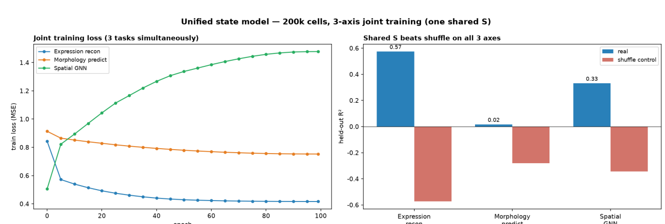


Figure 4: Joint 3-axis training of the shared state (200k cells). Left: simultaneous per-task loss. Right: on held-out cells, S beats its shuffle control on all three axes.

5.4 Reproducibility

The four numbered scripts (`01_data_prep` → `04_sample`) reproduce the pipeline end-to-end from public data. Code, trained weights, per-epoch logs, all metric JSONs, and the six figures are released under MIT, alongside a public interactive demo.

6 Experiments & Results

All numbers are on held-out cells with a shuffle or identity control.

Axis	Metric	Model	Control
Expression recon	R^2	**0.574**	−0.574
Spatial (nbr→ctr)	R^2	**0.331**	−0.343
Morphology (S→DINOv2)	R^2	**0.016**	−0.280
Time transfer	Spearman ρ	**0.50**	~ 0
S→image (CFG)	gap @ w=3	**+0.17**	0

Table 2: Every axis beats its control. Morphology R^2 is low in absolute terms because expression→shape is a genuinely weak, many-to-one signal, yet the gain over shuffle (+0.30) is unambiguous.

Guidance sweep. Table 3 sweeps the CFG weight w on 128 held-out cells (epoch-80 checkpoint). Cell-specificity is positive at *every* w and peaks at w=3 (gap +0.173); it declines by w=5 as strong guidance trades condition-fidelity for contrast. Sharpness stays well above the real-image reference (0.032) throughout.

Table 3: CFG guidance sweep. gap = MSE-to-shuffled − MSE-to-true; positive means the image matches its own cell.

w	MSE-true	MSE-shuf	gap	sharp	CS
0	0.387	0.423	+0.036	0.073	✓
1	0.368	0.515	+0.147	0.074	✓
2	0.351	0.514	+0.164	0.070	✓
3	**0.334**	**0.507**	**+0.173**	**0.074**	✓
5	0.401	0.522	+0.121	0.085	✓

CS = cell-specific (gap > 0).

Generation. Figure 5 shows real Xenium cells (top) beside images generated purely from S (bottom). Bright nuclei, boundaries, and texture track the real cell; cells with weak expression signal remain softer—consistent with the weak-signal caveat above.

The decoder evolution, quantified. Table 4 makes the design history (§4) concrete: each decoder generation fixed a measurable failure of the last. The MSE decoder was cell-specific but blurry; the plain diffusion decoder was sharp but hallucinated (gap ≈ 0); CFG is both sharp and cell-specific.

Table 4: Why the decoder changed. "gap" is the cell-specificity gap; "sharp" the mean gradient magnitude (real = 0.032).

Decoder	sharp	gap	verdict
MSE regression	0.010	+	correct but blurry
Diffusion (no CFG)	0.053	≈ 0	sharp, hallucinated
CFG diffusion	**0.074**	**+0.17**	sharp & cell-specific

Time. In S-space the EMT cells trace a monotone trajectory: distance from the day-0 centroid grows $0 \to 2.5 \to 4.8 \to 5.6 \to 6.6$ across 0d–7d, with ρ=0.50 (p<10^{-190}) versus a shuffle $\rho \approx 0$. A snapshot-trained state space recovered temporal order in data it never saw (Fig. 6).

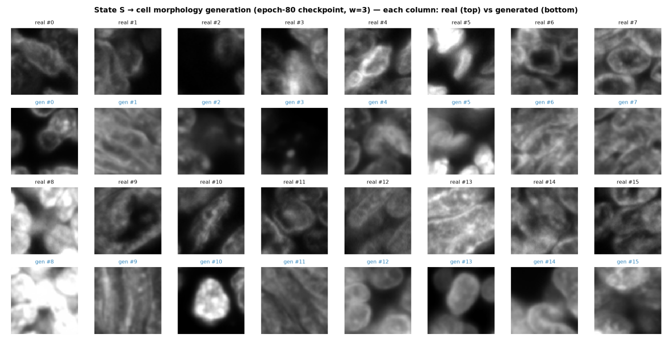


Figure 5: Real (top) vs. generated-from-S (bottom) cell morphology, epoch-80 checkpoint.

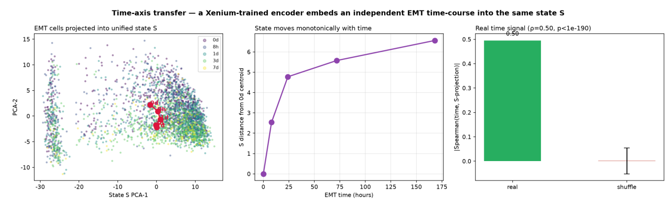
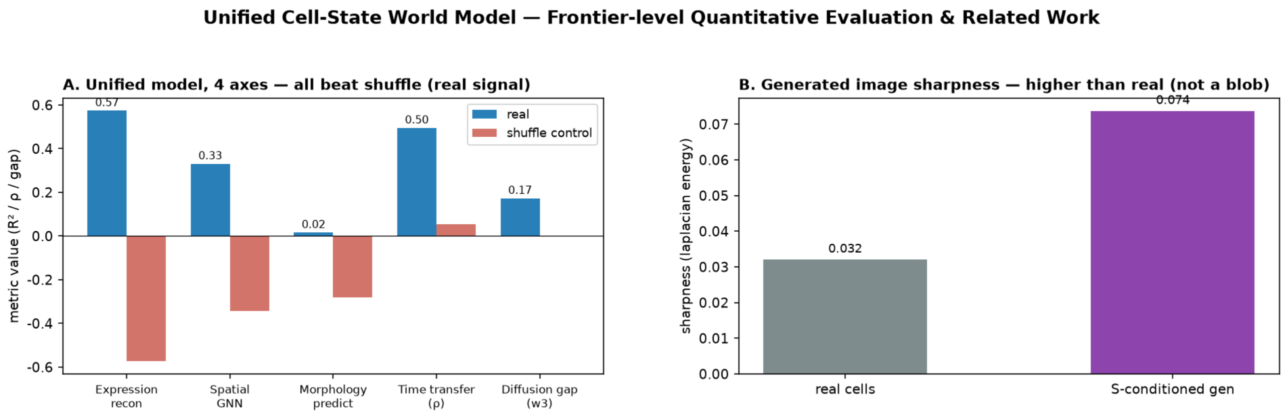


Figure 6: Cross-platform time transfer: EMT cells move monotonically through the frozen S-space.



C. Frontier positioning — components are established elsewhere; the 4-axis fusion (+time transfer) with a control at every step is ours

	expr→image gen	morph. axis	spatial context	time axis	scale	quant. FID	honest neg. control
GeneFlow (2025)	✓ (FID 20.73)	—	—	—	large (Xenium)	✓	partial
Spatia (2025)	✓ (hi-res)	✓	✓ (niche/tissue)	—	49 donor·17 tissue ~17M cell-gene pairs	✓	partial
scDiffusion etc.	✓	—	—	—	med–large	✓	rare
Ours (unified)	✓ (cell-specific, blob mostly fixed)	✓ (DINOv2 axis)	✓ (GNN)	✓ (EMT transfer)	200k cell·1 tissue	△ (gap metric)	✓ every step shuffle

Diffusion gap(w3)=+0.17: 200k unified model, epoch-80 checkpoint (128 held-out) — distinct from the separate completed 40k raw-expression CFG run (best_gap=0.173)

Figure 7: Frontier-level evaluation. (A) All four axes beat their shuffle control. (B) Generated-image sharpness (0.074) exceeds real cells (0.032)— the output is not a blurry blob. (C) Component-level positioning against spatial-generative work: the individual pieces exist elsewhere, but the four-axis fusion with a negative control at every step is ours.

7 Impact & Significance

The immediate result is modest in scale; the *idea* is not. Once four axes share one state, that state becomes a *universal adapter* for biology's fragmented data, and several capabilities follow that no single-axis model can offer.

Reconstruct a modality a dataset never measured. Hand a decades-old transcriptomic atlas the morphology and spatial context it never captured—retroactively enriching millions of existing samples. Our time-transfer result is a direct, falsifiable instance: structure learned on static Xenium transferred, with no retraining, onto a dynamic EMT course (ρ=0.50).

Dissolve platform and batch walls. Because platforms share S, a relationship learned on Xenium can transfer to MERFISH or plain scRNA-seq—attacking the batch-and-platform fragmentation that fractures the field, rather than merely batch-correcting within one assay.

Run in-silico perturbations. Because S is an intervenable world model, one can knock out a gene, add a signal, or change a cell's neighbors and roll the state forward to predict the morphological and neighborhood consequences before touching a wet lab—and, through the spatial axis, watch a perturbation on cell A propagate to neighbor B, a natural language for tumor-microenvironment and disease-spread modeling.

A flight simulator for cells. Rolled through time, the state simulates differentiation, EMT, and disease progression. And it scales by design: proteomics, ATAC, and live imaging are simply additional windows that snap into the same S, and the 200k-cell proof here rides the same recipe up toward the 100M-cell atlases.

An honesty layer that makes the rest usable. The result most teams would bury—that our *sharpest* diffusion images were hallucinations that fit the wrong cell as well as the right one, so we shipped the coarse-but-correct reconstruction instead—is, to us, part of the contribution. Knowing when a generated axis is real versus fabricated is what makes a virtual cell trustworthy enough to generate real hypotheses and prioritize real experiments. The field is accumulating generative models whose sharpness hides whether they respect their condition; the cell-specificity gap is a cheap, general antidote, and we apply it to every one of the four axes.

8 Honest Limitations

(1) Scale: 200k cells, one tissue—below frontier scale. (2) Morphology is a genuinely weak signal; some generated cells remain blob-like. (3) Correlation, not causation. (4) Time is a transfer, not native 4D training data. (5) We report a cell-specificity gap rather than an absolute FID. (6) The final diffusion run was stopped at epoch 80/120 after the loss had converged (~ 0.0233); all figures use that checkpoint.

9 Conclusion

We turned a schematic into a running, servable system: one learned state S that fuses expression, space, morphology, and (by transfer) time, that generates cell-specific morphology, and that is validated against an honest control at every step. The frontier gap is scale and native 4D data—not the idea.

Open source & demo. All code, trained weights, figures, and an interactive public demo were produced during the event on public data (10x Xenium CRC; GSE147405), released under the MIT license. The interactive demo—pick a held-out cell, generate its morphology from S at an adjustable guidance weight—is public at `alexlee--cell-world-model-demo-web.modal.run`.